GOLD

Gold Monetary, Geopolitical & Supply Risk Intelligence Model

A quantitative decision-support study of monetary transmission, geopolitical pressure, market stress, and mining-equity sensitivity.

Model version	Data mode	Analysis date
0.1.1	LIVE	2026-08-04

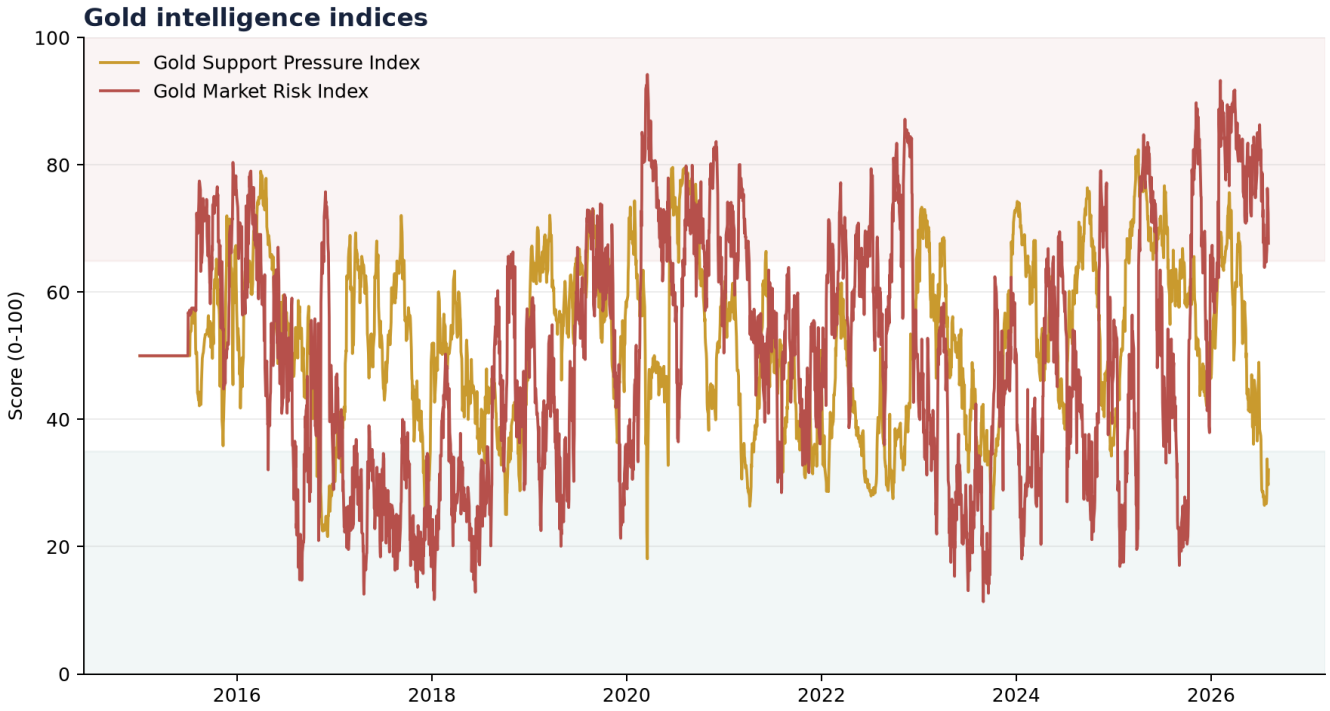
This report uses live public data.

Prepared as an analytical portfolio project by Jason Keen

Executive assessment

Index	Score	Level	As of
Gold Support Pressure Index	32.1	Low	2026-08-04
Gold Market Risk Index	67.7	High	2026-08-04

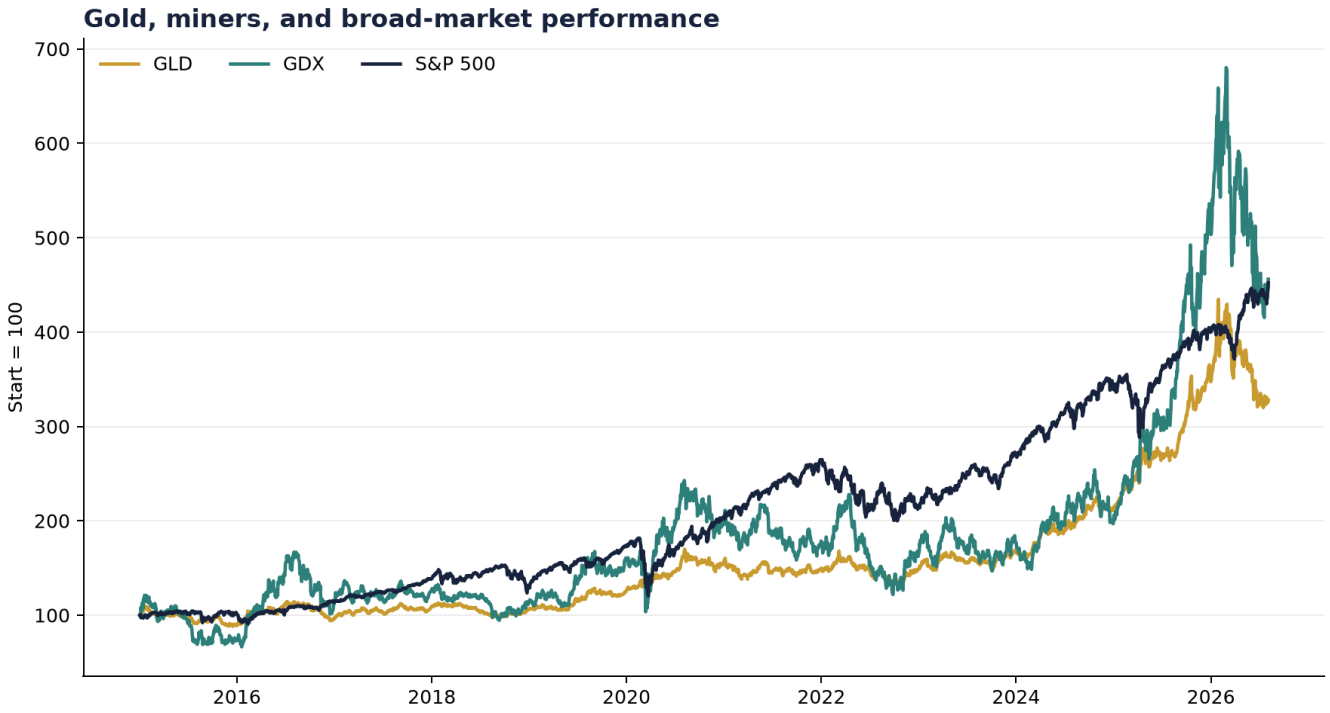
Current conditions produce a Gold Support Pressure score of **32.1** and a Gold Market Risk score of **67.7**. The separation is intentional: conditions that support gold prices can coexist with elevated volatility, crowding, or miner-specific risk.



Research design

The core test asks whether geopolitical pressure has an independent association with gold returns after controlling for real yields, the broad U.S. dollar, VIX, inflation expectations, and the S&P 500. A second set of models estimates how strongly the gold-mining ETF and major producers transmit changes in gold and the broader market.

Market performance

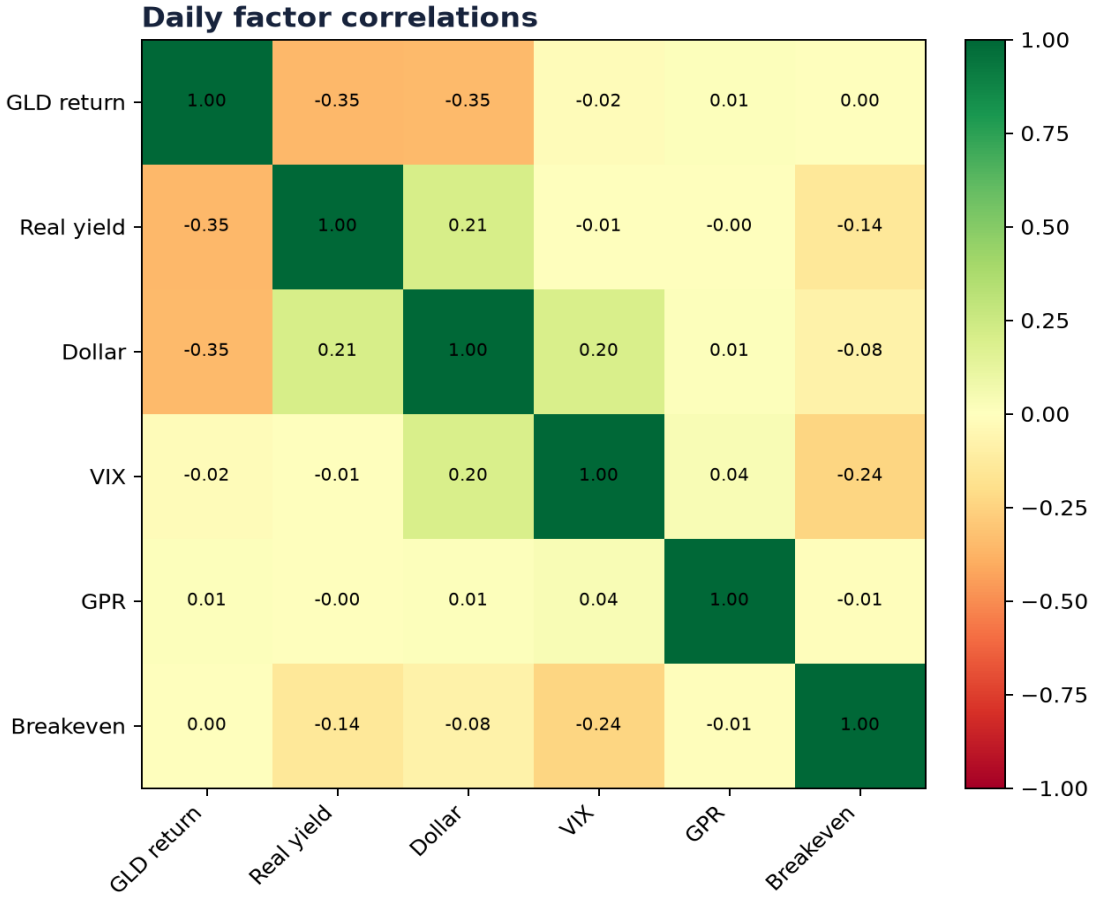


Asset	CAGR	Volatility	Sharpe	Max drawdown	Gold beta
GLD	10.8%	16.1%	0.51	-26.4%	1.00
GDX	14.0%	39.2%	0.28	-49.8%	1.87
Newmont	17.5%	37.5%	0.38	-62.4%	1.54
Barrick	24.1%	45.2%	0.43	-62.9%	0.66
Agnico Eagle	18.0%	39.4%	0.37	-54.2%	1.73
AngloGold Ashanti	23.3%	51.7%	0.37	-67.9%	2.21

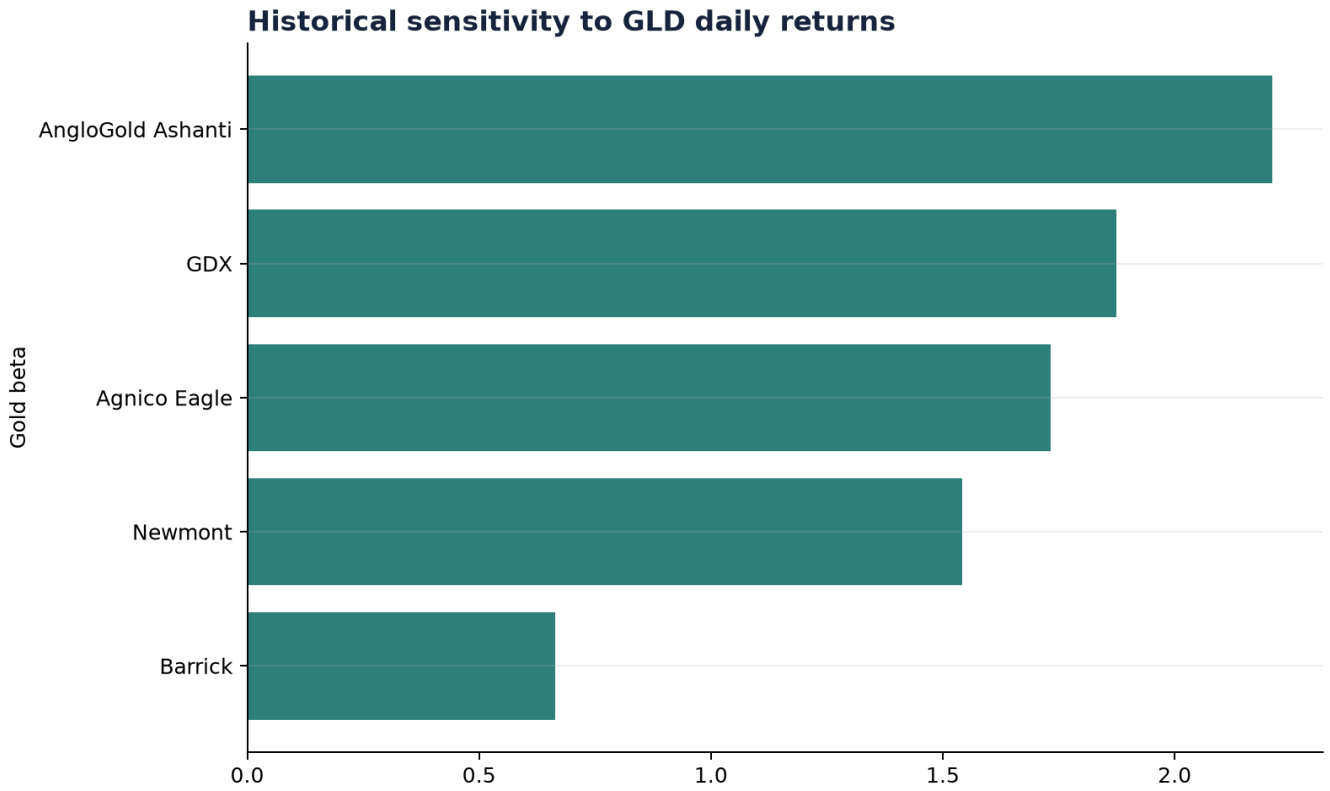
Factor transmission

Factor	Coefficient	t-statistic	R-squared	N
Real-yield change	-0.0799	-4.47	0.403	139
Dollar return	-1.3086	-5.67	0.403	139
VIX change	0.0282	1.42	0.403	139
GPR change	0.0158	1.91	0.403	139
Breakeven change	-0.0030	-0.11	0.403	139
S&P 500 return	-0.2123	-1.52	0.403	139

Coefficient signs and t-statistics describe conditional historical relationships. They do not establish that a factor caused a gold-price movement. Collinearity, regime changes, and omitted variables remain material limitations.

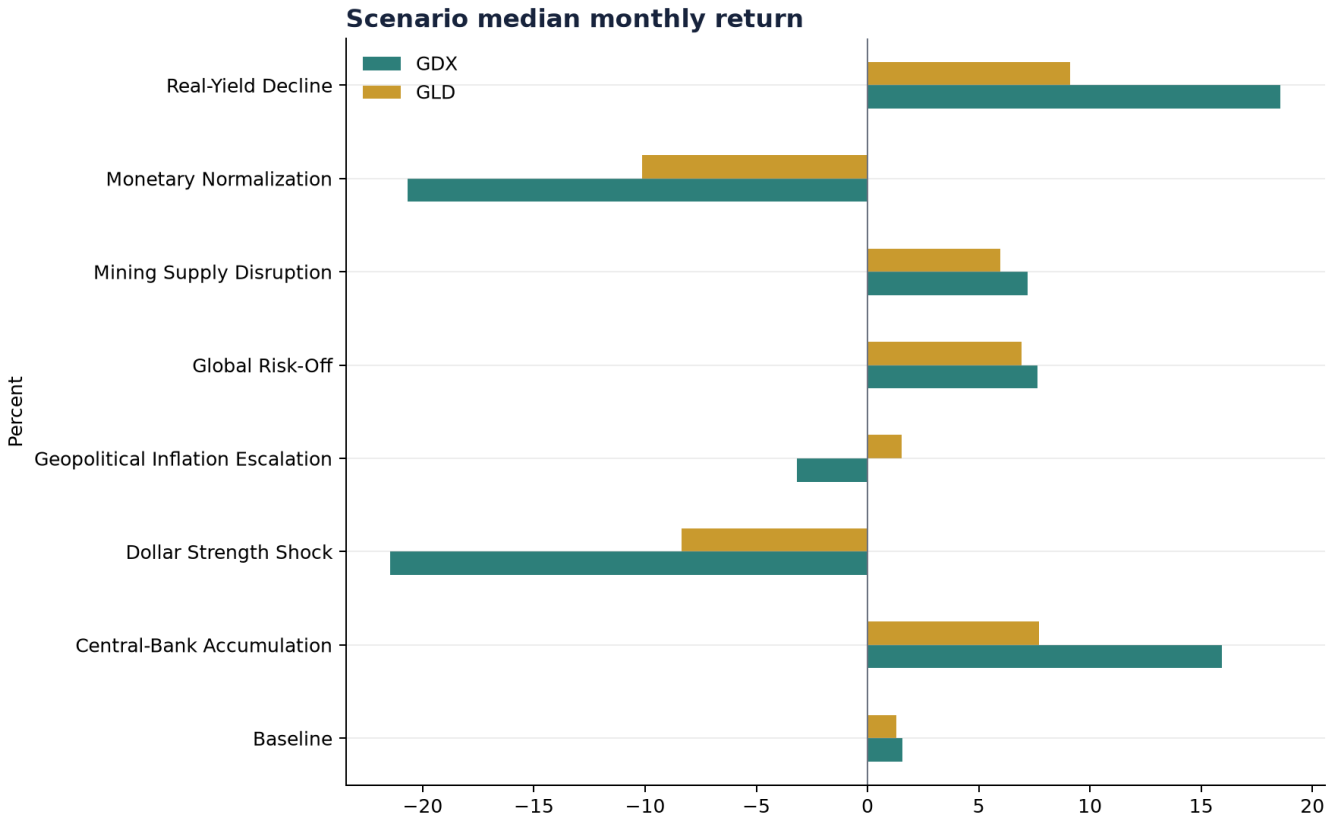


Mining-equity sensitivity



A gold beta above one indicates that the equity historically moved more than GLD in the same direction on a daily basis. Company performance also reflects operating costs, jurisdiction, capital allocation, currency exposure, hedging, reserve quality, and market beta.

Scenario intelligence



Scenario	Median	5th pct.	95th pct.	P(positive)
Real-Yield Decline	9.1%	3.5%	14.9%	100%
Central-Bank Accumulation	7.7%	2.0%	13.5%	99%
Global Risk-Off	6.9%	1.5%	12.8%	98%
Mining Supply Disruption	6.0%	0.1%	11.4%	95%
Geopolitical Inflation Escalation	1.5%	-4.3%	7.2%	67%
Baseline	1.3%	-4.6%	6.9%	64%
Dollar Strength Shock	-8.3%	-14.1%	-2.7%	1%
Monetary Normalization	-10.1%	-15.9%	-4.6%	0%

Data governance and limitations

Dataset	Status	Observations	Latest date
Gold Futures	live	2912	2026-08-04
GLD	live	2913	2026-08-04
GDX	live	2913	2026-08-04
Newmont	live	2913	2026-08-04
Barrick	live	2913	2026-08-04
Agnico Eagle	live	2913	2026-08-04
AngloGold Ashanti	live	2913	2026-08-04
S&P 500	live	2913	2026-08-04
Real Yield	live	2896	2026-07-31
Broad Dollar	live	2887	2026-07-31
VIX	live	2944	2026-08-03
Breakeven Inflation	live	2897	2026-08-03
GPR	live	4230	2026-07-31

Known limitations

- Optional central-bank demand is neutral when no reviewed, reusable time series is supplied.
- CFTC positioning and country-level supply concentration are documented extensions and are neutral in version 0.1.1.
- Yahoo price data can be revised and is not an institutional market-data feed.
- The factor models assume stable linear relationships and may fail during structural breaks.
- Scenario intervals reflect model and residual uncertainty, not the full range of possible outcomes.
- This research is not investment advice and should not be used as a standalone trading signal.

Central-bank component: neutral fallback: no reviewed series

Positioning/supply component: neutral fallback: extension not yet connected